Sahaj Bhadja

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Summary

Software engineer with 2+ years supporting **platform integrations**, **automation-first operations**, and **production reliability** across Python-based workflows, CI/CD pipelines, cloud services, and secure data-transfer systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Built **Python-driven automation services** and operational interfaces for 500+ users, reducing processing latency from **1.2s to 350ms** while improving platform observability and supportability.
- Implemented **fault-tolerant orchestration** for 5K+ monthly jobs with validation and rollback safeguards, strengthening production stability and incident response readiness.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed **workflow-oriented full-stack features** handling 10K+ daily requests, translating requirements into maintainable APIs and UI paths with clear deployment and support ownership.
- Improved **CI/CD and release automation** through scripted checks and defect triage, reducing turnaround by **60%** while preserving service reliability under frequent updates.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built **large-scale validation pipelines** over 100K+ trajectory logs, enabling reproducible testing and robust troubleshooting for high-complexity software behaviors.
- Produced **technical diagnostics and runbooks** that improved cross-team issue communication and accelerated root-cause resolution in fast-moving environments.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered **API and ETL integration workflows** with Python/Java, AWS Glue, and S3, improving response time by **30%** and reliability of distributed data transfers.
- Maintained **20+ automated unit/integration suites** with Jenkins, PyTest, and Mockito on Linux/Windows, boosting pipeline efficiency by **68%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Implemented **MERN + WebSocket workflows** for eight business paths and 2M+ synchronized records, improving consistency and monitoring across high-volume operational traffic.
- Automated **containerized deployment pipelines** with Docker/Jenkins, reducing release time by **50%** and improving repeatability across environments.

Skills

Languages: **Python**, **PowerShell**, Java, SQL, JavaScript/TypeScript, Bash
Platform Ops: production support, incident response, on-call readiness, platform maintenance, DR test mindset
Integration: managed file transfer concepts (SFTP/FTPS/S3), workload scheduling fundamentals, certificate lifecycle basics
Cloud & IaC: AWS services, AWS CDK (Python), GitHub Actions, Terraform/Ansible familiarity
Tooling: Git/GitHub, ServiceNow workflows, PowerBI reporting, Agile pairing/mobbing collaboration

Projects

Automated MFT Operations Toolkit | *Python, PowerShell, AWS, GitHub Actions*
- Built a **self-service operations toolkit** for secure file-transfer checks, certificate validation, and job health reporting, reducing manual support effort and improving audit readiness.

Cloud Transfer Scheduler Migration | *AWS CDK (Python), S3, Lambda, DynamoDB*
- Developed a **cloud-native scheduling prototype** for migration from legacy batch workflows, improving deployment consistency and reducing environment drift through Infrastructure as Code.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Implemented a **containerized REST processing pipeline** with automated CI/CD checks, improving reliability and operational continuity for production-scale data workloads.